

CH6.2 條件機率、獨立事件與貝氏定理

$$1. (1) \frac{1}{15} \quad (2) \frac{7}{30} \quad (3) \frac{1}{3} \quad (4) \frac{3}{10} \quad 2. (1) \frac{4}{15} \quad (2) \frac{44}{455}$$

$$3. (1) \frac{1}{6} \quad (2) \frac{1}{4} \quad (3) \frac{2}{3} \quad 4. (1) \frac{19}{50} \quad (2) \frac{4}{19} \quad 5. (1) \frac{2}{3} \quad (2) \frac{50}{81}$$

$$6.8 \quad 7. (1) 0.043 \quad (2) \frac{12}{43} \quad 8. \frac{21}{31} \quad 9. \frac{1}{20}$$

1. (1) 甲抽中有獎籤之機率為 $P(A) = \frac{3}{10}$ ，甲抽中後剩下 9 支籤，其中有獎籤有 2 支

$\therefore$ 此時乙抽到有獎籤之機率為 $P(B | A) = \frac{2}{9}$ ，故所求即 $P(A \cap B) = P(A) \cdot$

$$P(B | A) = \frac{3}{10} \cdot \frac{2}{9} = \frac{1}{15}$$

(2) 甲沒抽中有獎籤之機率為 $P(A') = \frac{7}{10}$

甲抽過後剩下 9 支籤，其中有獎籤 3 支

$\therefore$ 此時乙抽到有獎籤之機率為 $P(B | A') = \frac{3}{9}$ ，

故所求即 $P(A' \cap B) = P(A') \cdot P(B | A')$

$$= \frac{7}{10} \cdot \frac{3}{9} = \frac{7}{30}$$

(3) 所求即 $P(B | A') = \frac{3}{9} = \frac{1}{3}$

(4) 由(1)知甲抽中有獎籤之機率為 $P(A) = \frac{3}{10}$ 由(1)(2)知乙抽中有獎籤之機率為

$$\begin{aligned} P(B) &= P(B \cap A) + P(B \cap A') = P(A)P(B | A) + P(A')P(B | A') = \frac{1}{15} + \frac{7}{30} \\ &= \frac{9}{30} = \frac{3}{10} \end{aligned}$$

2. (1) 每次取到不良品之機率都是 $\frac{4}{15}$

$$\therefore P(\text{第三次 / 賽不良品}) = \frac{4}{15}$$

(2) 前兩次恰取到一個良品、一個不良品:

$$C_1^2 \times \frac{4}{15} \times \frac{3}{14} \times \frac{11}{13} = \frac{44}{455}$$

$$3. P(A \cup B) = P(A) + P(B) - P(A \cap B) \cdots \textcircled{1}$$

$$(1) \text{由} \textcircled{1} \quad \frac{1}{2} = \frac{1}{3} + P(B) - 0 \quad \therefore P(B) = \frac{1}{6}$$

$$(2) \text{由} \textcircled{1} \quad \frac{1}{2} = \frac{1}{3} + P(B) - P(A)P(B)$$

$$= \frac{1}{3} + P(B) - \frac{1}{3}P(B)$$

$$\therefore \frac{1}{6} = \frac{2}{3}P(B) \Rightarrow P(B) = \frac{1}{4}$$

(3) A' , B' 亦為獨立

$$\therefore P(A' | B') = P(A') = 1 - \frac{1}{3} = \frac{2}{3}$$

4. (1) $P(\text{恰中二})$

$$\begin{aligned} &= 0.4 \times 0.5 \cdot (1 - 0.6) + 0.4(1 - 0.5) \times 0.6 + (1 - 0.4) \times 0.5 \times 0.6 \\ &= \frac{4}{10} \times \frac{1}{2} \times \frac{4}{10} + \frac{4}{10} \times \frac{1}{2} \times \frac{6}{10} + \frac{6}{10} \times \frac{1}{2} \times \frac{6}{10} = \frac{76}{200} = \frac{19}{50} \end{aligned}$$

$$(2) P(\text{甲, 乙命中} | \text{恰中二}) = \frac{\frac{\frac{4}{10} \times \frac{1}{2} \times \frac{4}{10}}{\frac{19}{50}}}{\frac{19}{50}} = \frac{4}{19}$$

$$5. (1) \quad \frac{2}{3} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{2}{3} = \frac{6}{9} = \frac{2}{3}$$

(2) 分第一局後 A 點 3 白或 2 白 1 黑或 1 白 2 黑討論:

$$\left(\frac{1}{3} \cdot \frac{1}{3}\right) \cdot \left(\frac{3}{3} \cdot \frac{2}{3}\right) + \frac{2}{3} \cdot \frac{2}{3} +$$

$$\left(\frac{2}{3} \cdot \frac{1}{3}\right) \cdot \left(\frac{2}{3} \cdot \frac{2}{3}\right) = \frac{6}{81} + \frac{36}{81} + \frac{8}{81} = \frac{50}{81}$$

6. 設至少要發射 n 發，則 $1 - P(\text{全不中}) > 0.98$

$$\Rightarrow 1 - (1 - 0.4)^n > 0.98 \Rightarrow (0.6)^n < 0.02$$

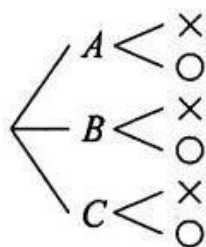
$$\Leftrightarrow \left(\frac{3}{5}\right)^n < \frac{1}{50} \Leftrightarrow \left(\frac{5}{3}\right)^n > 50 \Leftrightarrow n(\log 5 - \log 3) > \log 50$$

$$\Rightarrow n > \frac{\log 50}{\log 5 - \log 3} = \frac{1 + 0.6990}{0.6990 - 0.4771} = 7. \dots \dots$$

故至少要發射 8 發

7. (1) P (瑕疵品)

$$\begin{aligned} &= \frac{50}{100} \times \frac{5}{100} + \frac{30}{100} \times \frac{4}{100} + \frac{20}{100} \times \frac{3}{100} \\ &= \frac{25}{1000} + \frac{12}{1000} + \frac{6}{1000} = 0.043 \end{aligned}$$



(2) $P(B \mid \text{瑕疵品})$

$$= \frac{\frac{30}{100} \times \frac{4}{100}}{\frac{50}{100} \times \frac{5}{100} + \frac{30}{100} \times \frac{4}{100} + \frac{20}{100} \times \frac{5}{100}} = \frac{12}{43}$$

$$8. \frac{\frac{\frac{3}{8} \cdot \frac{7}{10} \cdot \frac{3}{5}}{\frac{3}{8} \cdot \frac{7}{10} \cdot \frac{3}{5} + \frac{5}{8} \cdot \frac{3}{10} \cdot \frac{2}{5}}}{\frac{3}{8} \cdot \frac{7}{10} \cdot \frac{3}{5} + \frac{5}{8} \cdot \frac{3}{10} \cdot \frac{2}{5}} = \frac{21}{31}$$

$$9. \text{令患癌之比例為 } x \Leftrightarrow \frac{(1-x) \cdot \frac{10}{100}}{\frac{90x}{100} + (1-x) \cdot \frac{10}{100}} = \frac{19}{28} \Rightarrow x = \frac{1}{20}$$